Untitled

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```
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.3.3
```

```
## Warning: package 'tidyr' was built under R version 4.3.3
```

```
## Warning: package 'readr' was built under R version 4.3.3
```

```
## Warning: package 'stringr' was built under R version 4.3.3
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✔ dplyr     1.1.2     ✔ readr     2.1.5
## ✔ forcats   1.0.0     ✔ stringr   1.5.1
## ✔ ggplot2   3.5.2     ✔ tibble    3.2.1
## ✔ lubridate 1.9.2     ✔ tidyr     1.3.1
## ✔ purrr     1.0.2
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
e errors
```

```
temperatures <- read_csv("https://wilkelab.org/SDS375/datasets/tempnormals.csv")  |>
  mutate(
    location = factor(
      location, levels = c("Death Valley", "Houston", "San Diego", "Chicago")
    )
  )  |>
  select(location, station_id, day_of_year, month, temperature)
```

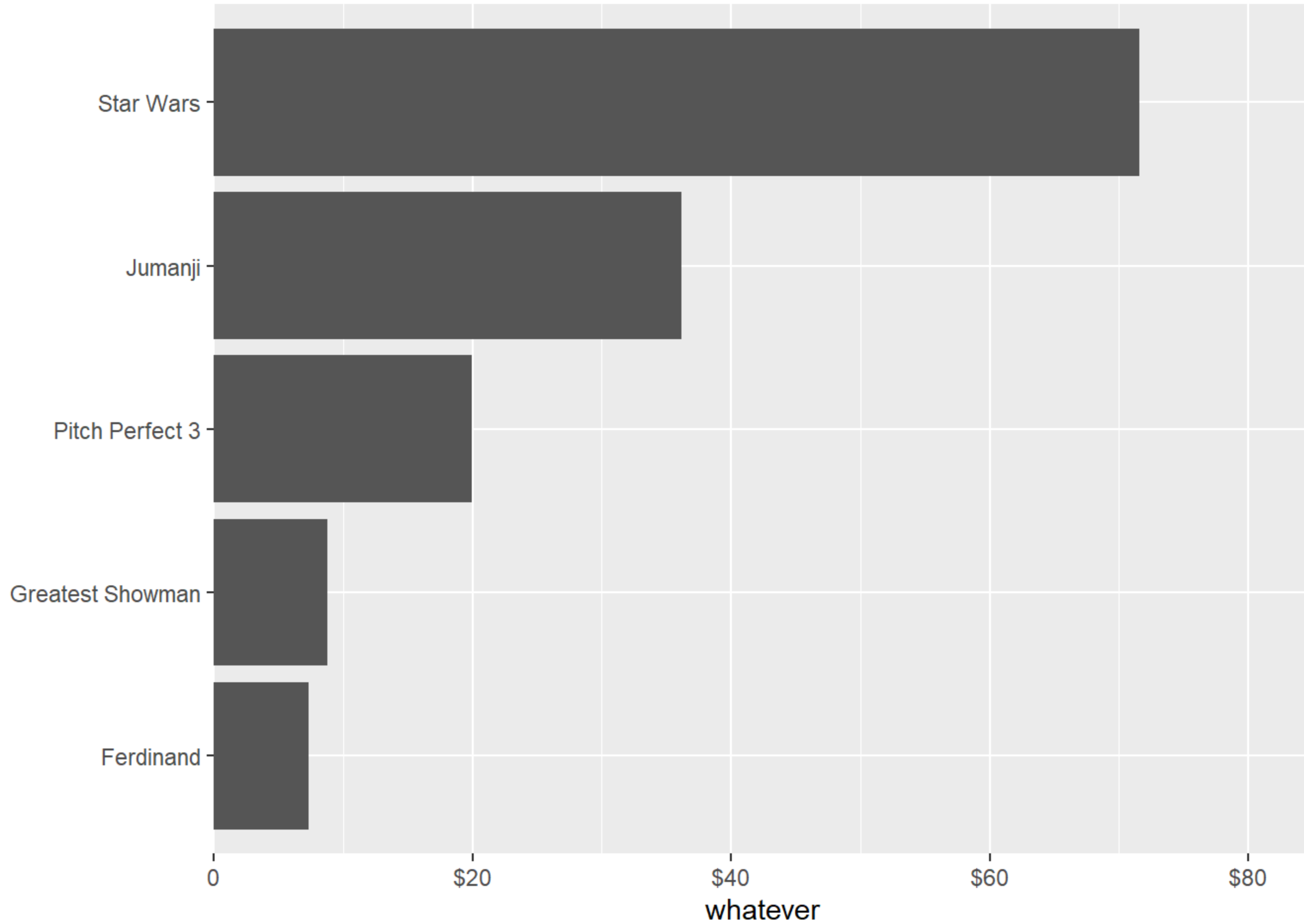
```
## Rows: 1464 Columns: 8
## ── Column specification ────────────────────────────────────────────────────────
## Delimiter: ","
## chr  (4): location, station_id, month_name, month
## dbl  (3): temperature, day, day_of_year
## date (1): date
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
boxoffice <- tibble(
  rank = 1:5,
  title = c("Star Wars", "Jumanji", "Pitch Perfect 3", "Greatest Showman", "Ferdinand"),
  amount = c(71.57, 36.17, 19.93, 8.81, 7.32) # million USD
)
```

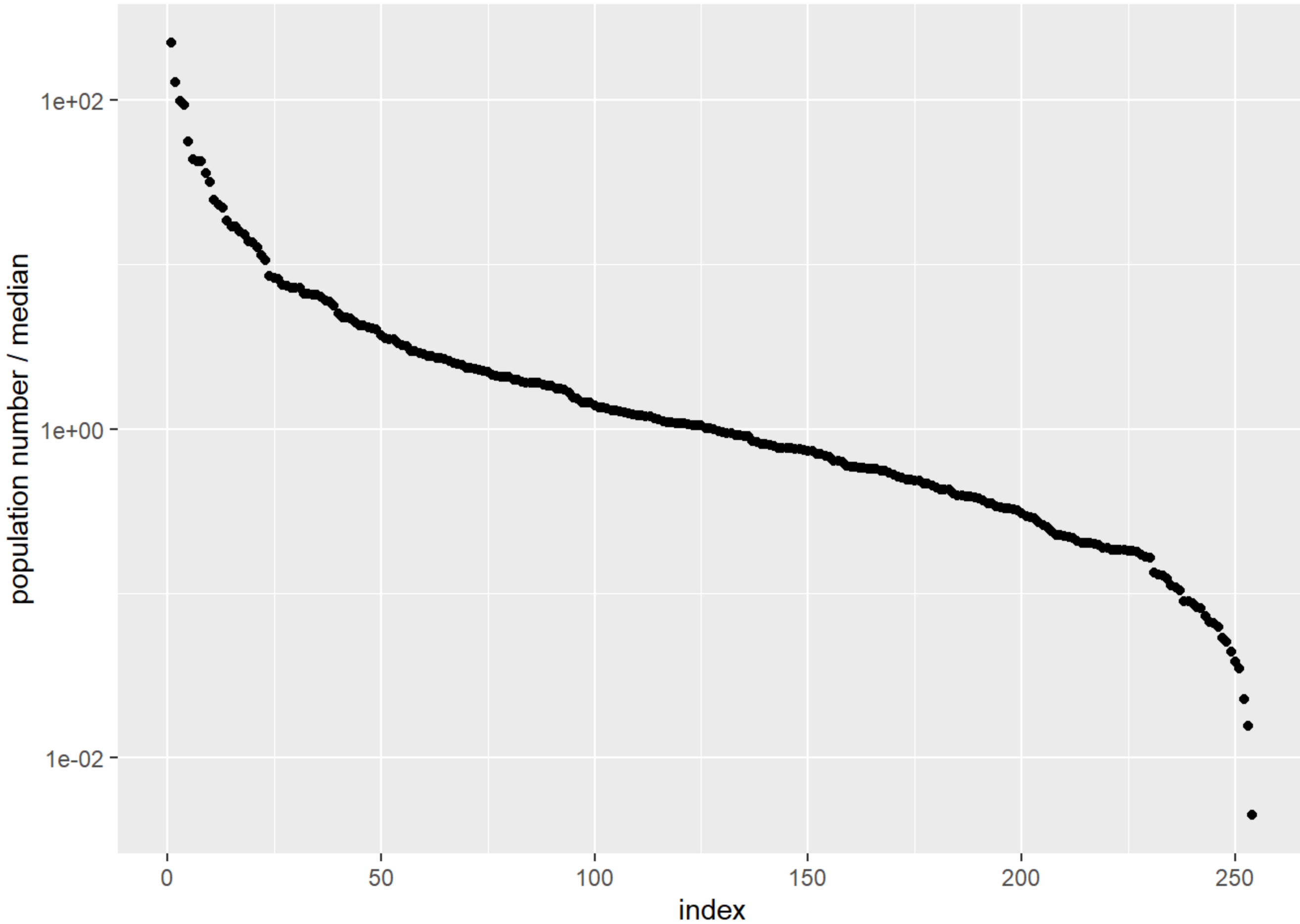
```
tx_counties <- read_csv("https://wilkelab.org/SDS375/datasets/US_census.csv") %>%
  filter(state == "Texas") %>%
  mutate(popratio = pop2010/median(pop2010)) %>%
  arrange(desc(popratio)) %>%
  mutate(index = 1:n())
```

```
## Rows: 3143 Columns: 53
## ── Column specification ────────────────────────────────────────────────────────
## Delimiter: ","
## chr  (3): state, name, FIPS
## dbl (50): pop2010, pop2000, age_under_5, age_under_18, age_over_65, female, ...
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
ggplot(boxoffice, aes(amount,fct_reorder(title,amount)))+
  geom_col()+
  scale_x_continuous(
    name = "whatever",
    limits = c(0,80),
    breaks = c(0,20,40,60,80),
    labels = c(0,"$20","$40","$60","$80"),
    expand = expansion(mult = c(0,0.06))
  )+
  scale_y_discrete(
    name = NULL
  )
```




```
ggplot(tx_counties, aes(index, popratio))+
  geom_point()+
  scale_y_log10(
    name = "population number / median",
    breaks = c(0.01,1,100)
  )
```




```
library(palmerpenguins)
```

```
## Warning: package 'palmerpenguins' was built under R version 4.3.3
```

```
library(ggthemes)
```

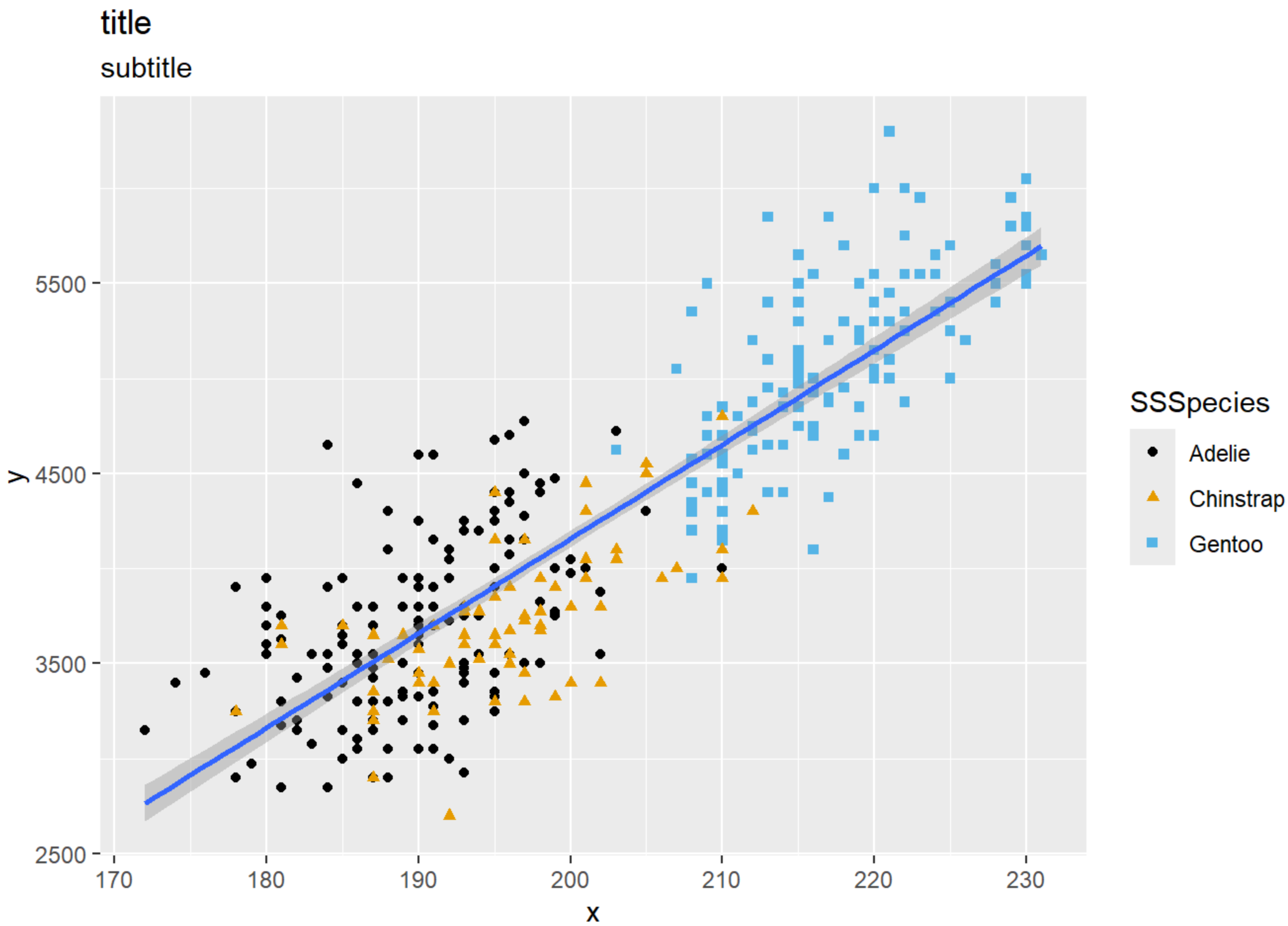
```
## Warning: package 'ggthemes' was built under R version 4.3.3
```

```
ggplot(penguins, aes(flipper_length_mm,body_mass_g))+
  geom_point(aes(color = species, shape = species))+
  geom_smooth(method = "lm")+
  labs(title = "title",
       subtitle = "subtitle",
       x = "x", y = "y",
       color = "SSSpecies", shape = "SSSpecies")+
  scale_color_colorblind()
```

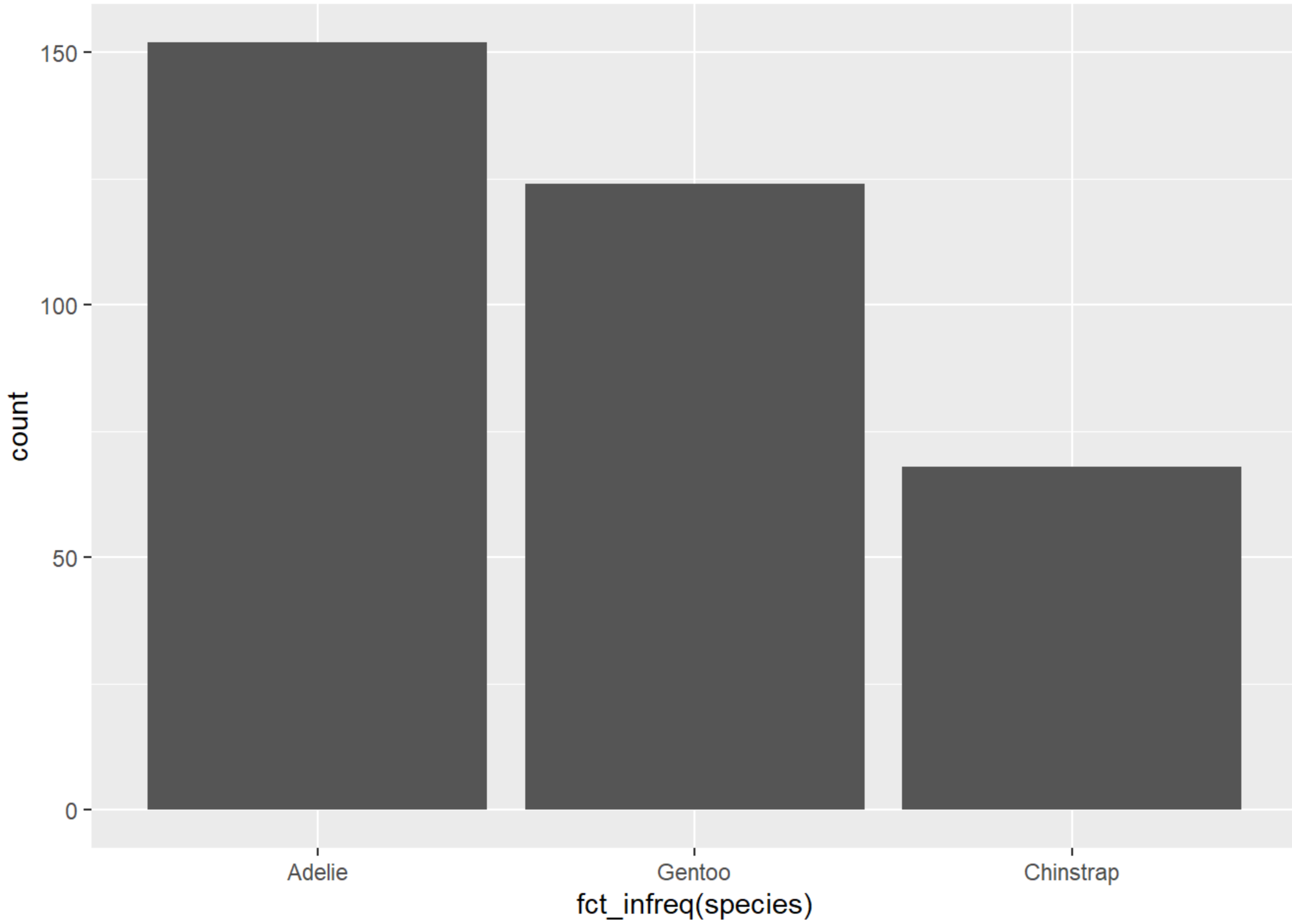
```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_smooth()`).
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

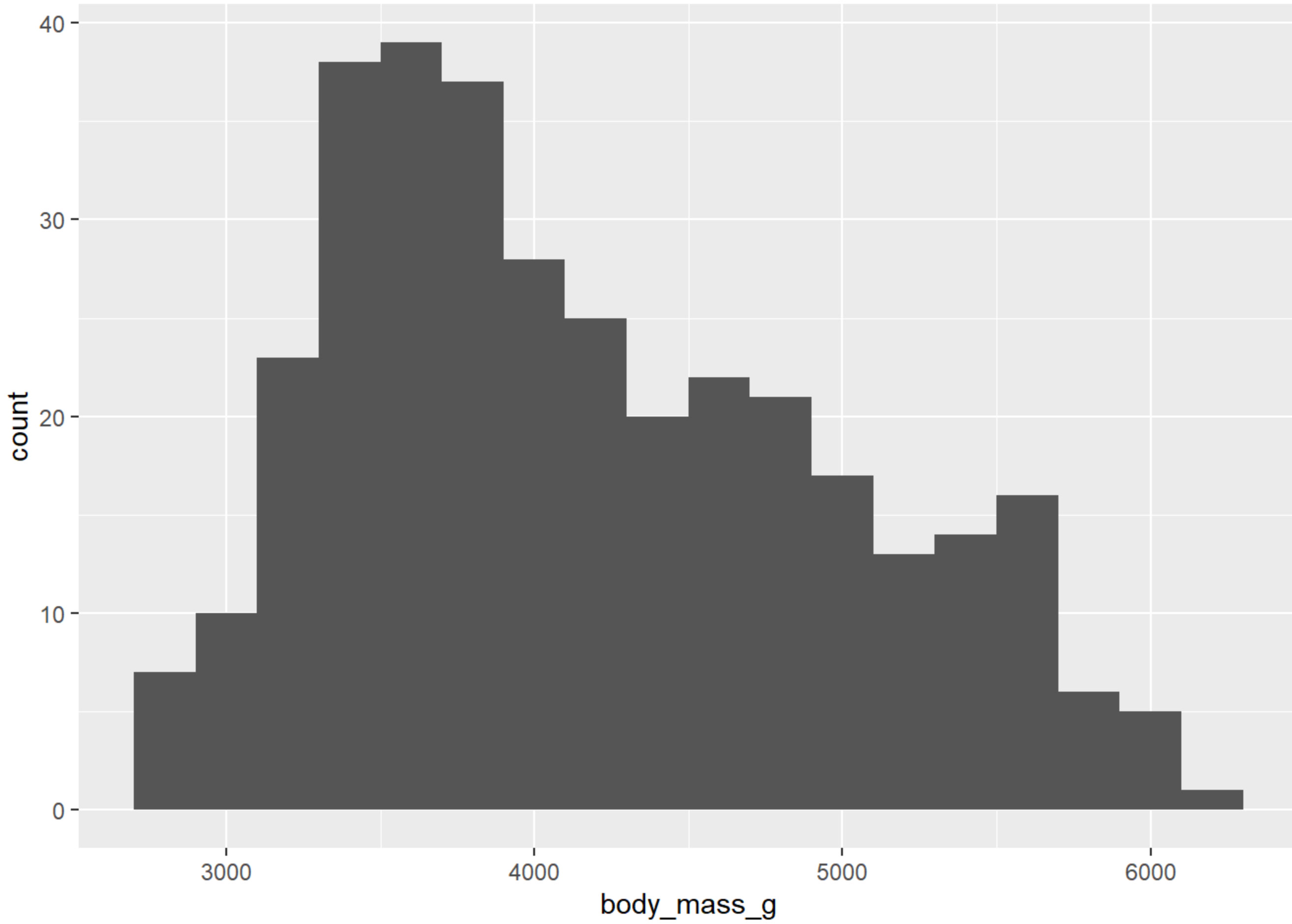



```
penguins |>
  ggplot(aes(fct_infreq(species)))+
  geom_bar()
```




```
penguins |>
  ggplot(aes(body_mass_g))+
  geom_histogram(binwidth = 200)
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

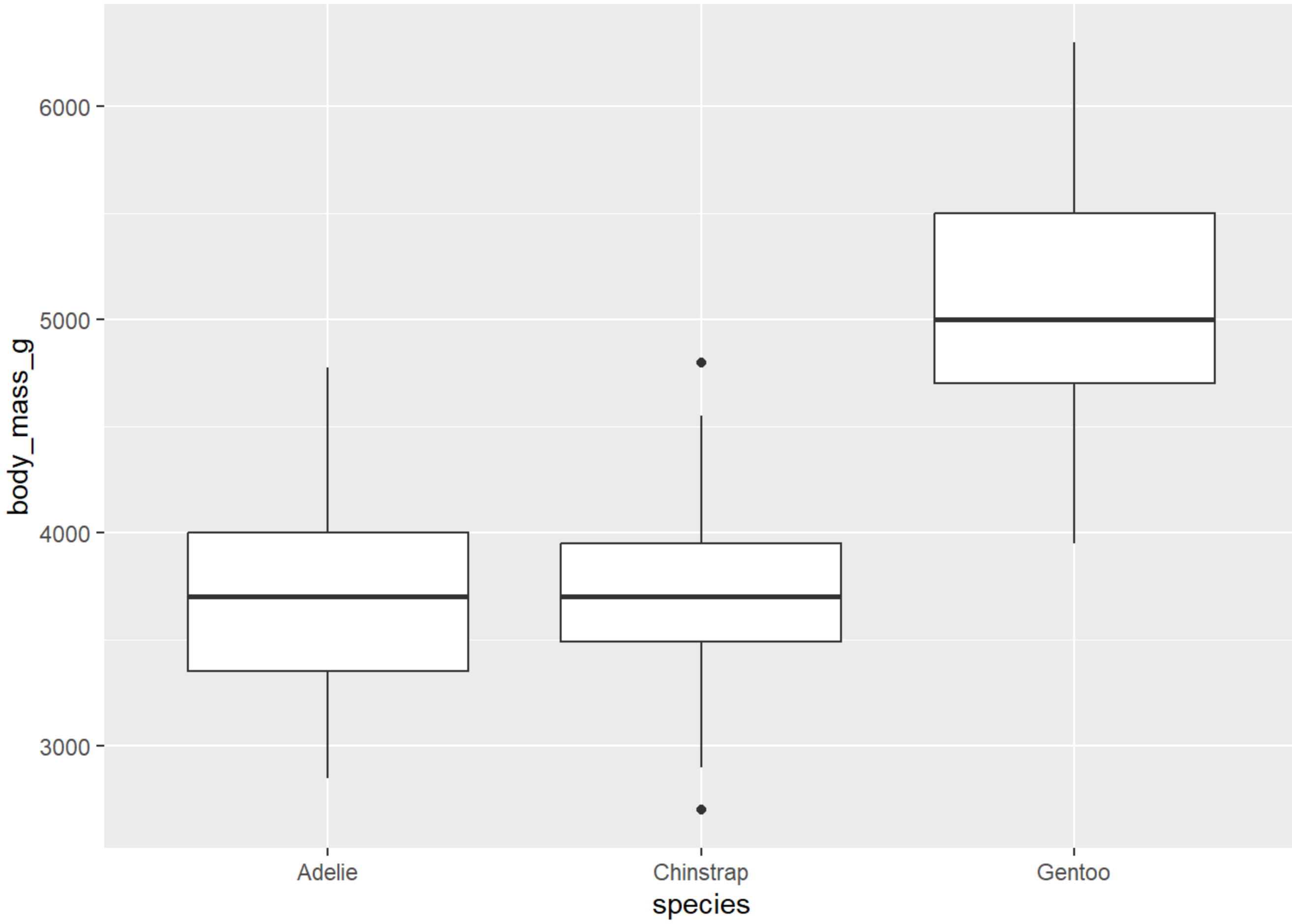



```
penguins |>
  ggplot(aes(body_mass_g))+
  geom_density(color = "red")
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_density()`).
```

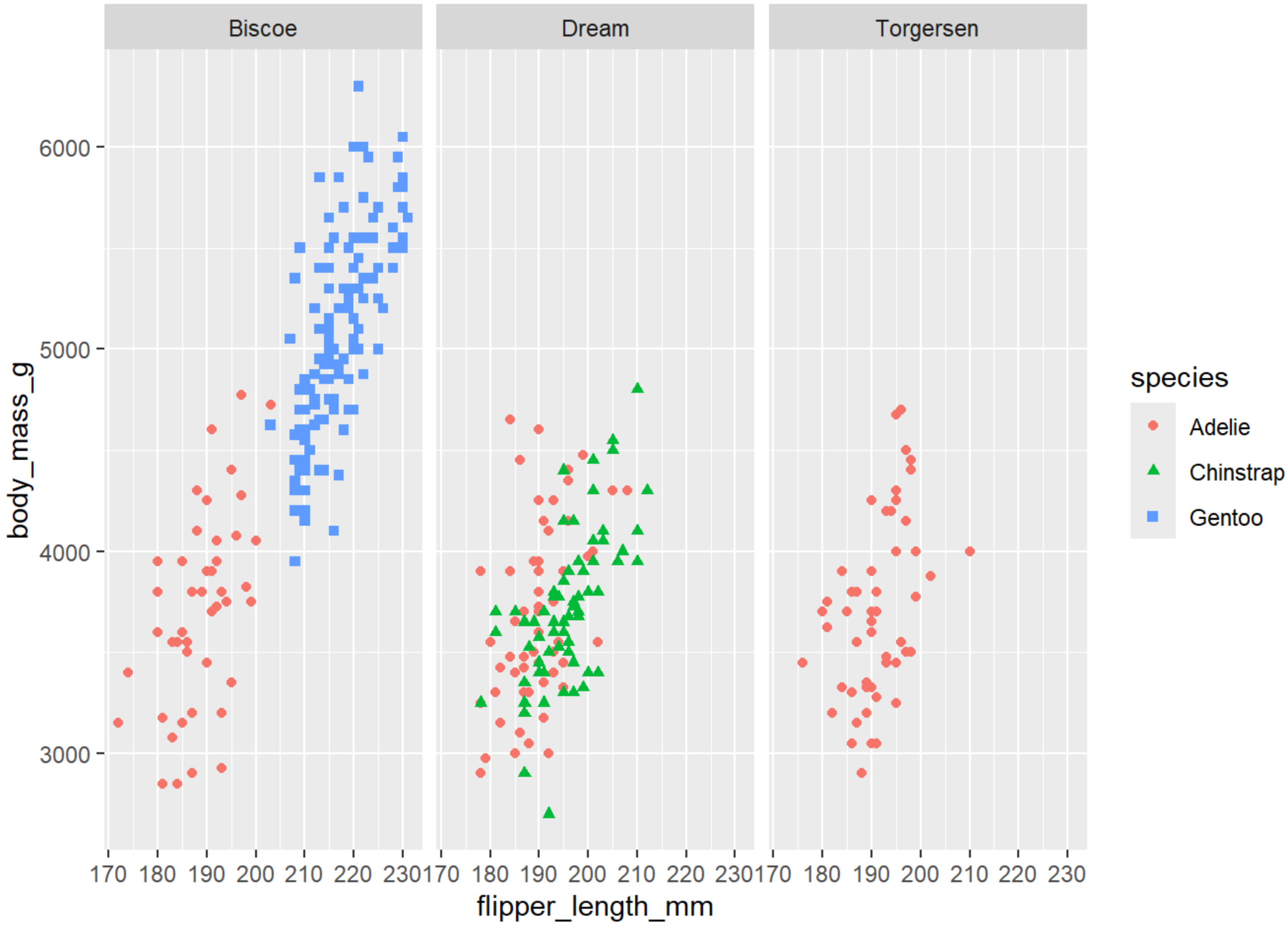

```
penguins |>
  ggplot(aes(species,body_mass_g))+
  geom_boxplot()
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```




```
penguins |>
  ggplot(aes(flipper_length_mm,body_mass_g)) +
  geom_point(aes(color = species, shape = species)) +
  facet_wrap(~island)
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
```


Biscoe
Dream
Torgersen
body_mass_g
6000
5000
4000
3000
170 180 190 200 210 220 230
flipper_length_mm
species
Adelie
Chinstrap
Gentoo